

**Charotar University of Science and Technology**  
**Faculty of Technology and Engineering**  
**Lesson Planning CEUE203 Object Oriented Programming (Theory)**

|                              |                     |  |                        |     |  |
|------------------------------|---------------------|--|------------------------|-----|--|
| <b>Academic Year :</b>       | 2026-27             |  | <b>Semester:</b>       | III |  |
| <b>Subject Coordinator :</b> | Prof Siddharth Shah |  | <b>Lec Hours/week:</b> | 2   |  |
| <b>Subject Teacher Name:</b> | Prof Sagar Patel    |  | <b>Div</b>             | I   |  |
| <b>Subject Code:</b>         | CEUE203             |  | <b>Subject Name:</b>   | OOP |  |

| Week | Unit number | Planned      |                           | Topics                                                                                                                                                                                                               | Actual                   |                                  | Remarks<br>(Activity, Lec swap, Combined lecture etc.) |
|------|-------------|--------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|----------------------------------|--------------------------------------------------------|
|      |             | Faculty Name | Planned Date [DD/MM/YYYY] |                                                                                                                                                                                                                      | Actual Date [DD/MM/YYYY] | Actual numbers of lectures taken |                                                        |
| 1    | Unit 1      | SMP          | 06/07/2026                | Java overview and architecture: history of Java, Java buzzwords (simple, object-oriented, robust, multithreaded, architecture-neutral, portable), JVM / JRE / JDK, bytecode and the 'write once, run anywhere' model |                          |                                  |                                                        |
| 1    | Unit 1      | SMP          | 07/07/2026                | Program structure and recent features: main method, compiling and running, switch expressions, records, enums and var                                                                                                |                          |                                  |                                                        |
| 2    | Unit 2      | SMP          | 13/07/2026                | Classes and objects: defining classes, creating objects with new, instance vs static variables and methods, access modifiers (public, private, protected, default)                                                   |                          |                                  |                                                        |
| 2    | Unit 2      | SMP          | 14/07/2026                | Constructors and keywords: default and parameterized constructors, constructor overloading, this and this() call, final with variables/methods/classes, instanceof                                                   |                          |                                  |                                                        |
| 3    | Unit 2      | SMP          | 20/07/2026                | Nested classes and the Object class: static nested, inner, method-local and anonymous classes; Object methods toString(), equals(), hashCode(), clone(); overriding toString() and equals()                          |                          |                                  |                                                        |
| 3    | Unit 2      | SMP          | 21/07/2026                | Memory, annotations and reflection: stack vs heap, object lifecycle and garbage collection; annotations and meta-annotations; custom annotations; reflection                                                         |                          |                                  |                                                        |
| 4    | Unit 3      | SMP          | 27/07/2026                | String handling: String class methods, StringBuffer and StringBuilder, string manipulation and immutability                                                                                                          |                          |                                  |                                                        |
| 4    | Unit 3      | SMP          | 28/07/2026                | Regular expressions and tokenization: Pattern and Matcher, common regex patterns, string tokenization with split()                                                                                                   |                          |                                  |                                                        |
| 5    | Unit 4      | SMP          | 03/08/2026                | Inheritance and polymorphism: extends, super keyword, dynamic method dispatch and runtime polymorphism, final in inheritance                                                                                         |                          |                                  |                                                        |
| 5    | Unit 4      | SMP          | 04/08/2026                | Abstraction: abstract classes and methods, upcasting and downcasting objects in inheritance                                                                                                                          |                          |                                  |                                                        |
| 6    | Unit 4      | SMP          | 10/08/2026                | Interfaces: declaring and implementing interfaces, Interface Segregation Principle, default, static and private interface methods                                                                                    |                          |                                  |                                                        |
| 6    | Unit 4      | SMP          | 11/08/2026                | Advanced interfaces: multiple inheritance via interfaces, functional interfaces and lambdas, marker interfaces                                                                                                       |                          |                                  |                                                        |
| 7    | Unit 4      | SMP          | 17/08/2026                | Packages and modules: creating and importing packages, access across packages, static imports, JAR files, Java Module System (modules, descriptors, services, service loaders)                                       |                          |                                  |                                                        |
| 7    | Unit 5      | SMP          | 18/08/2026                | Exception handling basics: checked vs unchecked exceptions, try-catch-finally, exception class hierarchy                                                                                                             |                          |                                  |                                                        |
| 8    | Unit 5      | SMP          | 24/08/2026                | Advanced exception handling: throw and throws, custom exceptions, try-with-resources, exception propagation                                                                                                          |                          |                                  |                                                        |
| 8    | Unit 6      | SMP          | 25/08/2026                | Introduction to multithreading: processes vs threads, life cycle of a thread, ways to create threads                                                                                                                 |                          |                                  |                                                        |
| 9    | Unit 6      | SMP          | 31/08/2026                | Creating threads: Thread class and Runnable interface, thread groups, start() vs run()                                                                                                                               |                          |                                  |                                                        |
| 9    | Unit 6      | SMP          | 01/09/2026                | Thread synchronization: synchronized methods and blocks, race conditions, inter-thread communication (wait/notify)                                                                                                   |                          |                                  |                                                        |
| 10   | Unit 6      | SMP          | 07/09/2026                | Thread coordination: thread priority and scheduling, deadlock and thread-safety                                                                                                                                      |                          |                                  |                                                        |
| 10   | Unit 6      | SMP          | 08/09/2026                | Executors and parallelism: Executor and ExecutorService for thread-pool management, Fork/Join framework                                                                                                              |                          |                                  |                                                        |
| 11   | Unit 7      | SMP          | 14/09/2026                | File handling with NIO.2: Path interface, Files class, file attributes                                                                                                                                               |                          |                                  |                                                        |
| 11   | Unit 7      | SMP          | 15/09/2026                | Streams and serialization: byte and character streams, object serialization and deserialization                                                                                                                      |                          |                                  |                                                        |
| 12   | Unit 7      | SMP          | 21/09/2026                | NIO fundamentals: Buffer, Channel and Selector                                                                                                                                                                       |                          |                                  |                                                        |
| 12   | Unit 7      | SMP          | 22/09/2026                | Directory operations: directory streams and file walking                                                                                                                                                             |                          |                                  |                                                        |
| 13   | Unit 8      | SMP          | 28/09/2026                | Collections and List: Collection Framework overview, Collection interface, List implementations (ArrayList, LinkedList, Vector, Stack)                                                                               |                          |                                  |                                                        |
| 13   | Unit 8      | SMP          | 29/09/2026                | Set: HashSet, LinkedHashSet, TreeSet and their use cases                                                                                                                                                             |                          |                                  |                                                        |
| 14   | Unit 8      | SMP          | 05/10/2026                | Map: HashMap, LinkedHashMap, TreeMap, Hashtable                                                                                                                                                                      |                          |                                  |                                                        |
| 14   | Unit 8      | SMP          | 06/10/2026                | Working with collections: iterating, sorting and searching, Comparable and Comparator, thread-safe collections                                                                                                       |                          |                                  |                                                        |
| 15   | Unit 8      | SMP          | 12/10/2026                | Generics: generic classes and methods, bounded type parameters                                                                                                                                                       |                          |                                  |                                                        |
| 15   | Unit 8      | SMP          | 13/10/2026                | Advanced generics and revision: wildcards, type erasure, course revision and integrated application walkthrough                                                                                                      |                          |                                  |                                                        |